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**THE EFFECT OF ACTIVE EMPLOYMENT PROGRAMS ON
YOUTH EMPLOYMENT: 2018 ADDITIONAL EMPLOYMENT
PROGRAM**

MASTER’S THESIS

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ABSTRACT

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Youth unemployment is an important problem faced by all world economies in recent years. It is clear that this problem will lead to more serious problems in the long run, especially in countries like Turkey with a high young population. On the other hand, unlike developed countries, another problem that Turkey struggles with is the high rate of informal employment. In this thesis, the effect of the *Additional Employment Incentive*, which came into force in 2018 in order to increase formal employment and prioritizes the employment of young men between the ages of 18-24, on the labor market is examined. Based on the age breakdown in the design of the program, the Difference of Differences Method and Regression Discontinuity Model were used. According to the findings obtained using the Difference of Differences Method, the program increased the employment probability of young men in the formal sector, while decreasing it in the informal sector. It was also observed that the program reduced real wages in the formal sector. Estimates made using the Regression Discontinuity Model, on the other hand, concluded that the probability of young men being employed in the formal sector increased only in the year the program entered into force.

Keywords: Youth Unemployment, Active Labor Employment Program, 2018 Additional Employment Program, Difference in Differences Method, Regression Discontinuity Design

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CHAPTER I: INTRODUCTION

Young people are considered one of the disadvantaged groups in the labor market. Compared to adults, their inadequate job search skills and lack of sufficient work experience lead to higher rates of unemployment and increased risk of employment in insecure jobs. Moreover, youth are known to be more vulnerable to cyclical fluctuations in the economy (O'Higgins, 1997). In their study testing Okun's Law for Turkey between 1989 and 2019, Coşar and Yavuz (2021) showed that the group most affected by economic fluctuations was individuals aged 15–24.

Active labor market programs are among the primary policies implemented by governments to combat unemployment and informal employment and to facilitate the integration of disadvantaged groups into the labor market. These programs are generally defined as policies that aim to align the supply and demand sides of the labor market and are classified into subcategories based on their mode of implementation. In this thesis, the impact of the 2018 Additional Employment Incentive on the labor market outcomes of young men is examined. The extension of the duration of insurance premium support for employers hiring males aged over 18 and under 25¹ allows this incentive to be classified as a wage subsidy-based active labor market program.

Examples of similar programs aimed at increasing employment by reducing hiring costs include tax reductions for employers hiring low-wage workers in France during the 1990s (Kramarz and Philippon, 2001; Crepon and Desplatz, 2002), the New Deal program in the UK targeting youth employment (Blundell et al., 2004; De Giorgi, 2005), Germany's Integration Allowance (EGZ) program designed to reintegrate older individuals into the labor market (Boockmann et al., 2012), and insurance premium support for employers hiring low-wage workers in Finland (Huttunen et al., 2013). Many studies evaluating active labor market programs have concluded that these programs increase employment among the target groups. However, some studies report no significant effects on labor market outcomes.

In Turkey, the 2008 Employment Support Program can be considered an example of a subsidy program aimed at promoting the employment of disadvantaged groups. The program provided insurance premium support to employers hiring women regardless of age and men aged over 18 and

¹Employers hiring individuals aged 25 and above are not eligible for insurance premium support.

under 29. Studies by Uysal (2013), Ayhan (2013), Balkan et al. (2016a), Dildar (2020), and İközler and Ökten (2020) investigated the program's impact on women's labor market outcomes. Balkan et al. (2016b) analyzed the program's effects on both women and young men, while Özdamar et al. (2021) focused solely on young men. Although most studies agree that the 2008 program increased women's employment, the findings regarding young men are more mixed.

This study analyzes the effect of the 2018 employment incentive package targeting young men aged 18–24 using data from the Household Labor Force Survey (HLFS) between 2016 and 2019. Both the Difference-in-Differences (DiD) method and the Regression Discontinuity Design (RDD) are employed. While estimating the program's impact using the DiD method, the influence of the 2008 program and other ongoing programs targeting similar groups was accounted for by selecting males aged 25–28 as the control group. Unlike previous studies analyzing the 2008 employment program, this thesis contributes to the literature by examining a different active labor market program's impact on the Turkish labor market. Furthermore, since the 2008 program was introduced during an economic crisis, its net effect is more difficult to isolate. Similarly, Cammeraat et al. (2017) found that a mandatory activation program introduced during the Great Recession (2007–2009) did not reduce the number of youth not in employment or education, attributing the ineffectiveness to the timing of the intervention. The fact that the 2018 Additional Employment Program was implemented during a non-crisis period is a distinct feature of this study.

To assess the impact of the 2018 Additional Employment Program using the DiD and RDD methods, the outcomes selected were employment status in the formal and informal sectors as a wage employee and hourly real wages. In the DiD estimation, control variables such as age, education level, and marital status were included to account for individual characteristics of young men. The years 2016–2017 were used as the pre-treatment period, and 2018–2019 as the post-treatment period. In the RDD approach, separate estimates were made for the years 2018 and 2019.

The DiD estimates first examined the overall impact of the program. The findings suggest that the 2018 Additional Employment Program increased the probability of formal sector employment among young men by 2 percentage points while reducing their real wages by 2%. In the informal sector, the program decreased the likelihood of employment by 0.8 points but had no significant effect on real wages. Following the general impact assessment, the year-specific effects of the program were evaluated. The results show that the program was more effective in its first year. Specifically, in 2018, the probability of formal sector employment increased by 2.5 points and

informal sector employment decreased by about 1 point. In 2019, the program increased formal sector employment probability by 1.6 points, but no statistically significant effect was found on informal employment or real wages. Further estimations by education level revealed that the program increased the likelihood of employment in the formal sector only for high school graduates. In the informal sector, it reduced the likelihood of employment among those with no school completion and university graduates. In terms of real wages, the program decreased real wages for primary and high school graduates in the formal sector, while having no significant effect on any group's wages in the informal sector. Estimations by age group showed that men aged 18–20 were the most affected by the program. A placebo test was conducted to verify the reliability of the estimates, and previous findings passed this test. Additional robustness checks using narrower age windows also yielded similar results. Lastly, taking advantage of the program's age eligibility criteria, RDD estimates were conducted using both linear and quadratic specifications.

The RDD results differ from the DiD estimates in some respects. According to RDD, the program increased the probability of employment in the formal sector. Although the effect on informal sector employment was negative, it was not statistically significant, and the placebo test for this variable failed. Furthermore, the program appears to have been effective only in the year it was introduced. Regarding real wages, the program had a significant effect only in the informal sector in 2019, but due to the small number of observations, the reliability of these estimates is questionable. The RDD results also proved to be highly sensitive to the chosen age intervals. Moreover, when control variables were added to the model, some differences in the estimates were observed.

In the second section of the thesis, previous studies examining the effects of active labor market policies in Turkey and globally are reviewed, highlighting the contribution of this thesis to the literature. The third section describes the structure of concurrent employment incentives in Turkey. The fourth section provides a theoretical background on wage subsidies. The fifth section introduces the dataset used and presents descriptive statistics. The sixth section explains the econometric strategy employed to estimate the causal effect of the 2018 employment program, separately for DiD and RDD methods. The seventh section presents the estimation results, and the eighth section provides interpretations and policy recommendations based on the findings.

CHAPTER II: LITERATURE REVIEW

Active labor market programs aim to improve the functioning of the labor market by enhancing its mobility and adaptability, and by facilitating the employment of the unemployed (Boeri & Van Ours, 2013). The functions of active labor market programs in the labor market include increasing economic output and welfare by enabling the unemployed to participate in employment or by raising individuals' human capital, preserving the size of the effective workforce by sustaining competition for existing jobs, supporting the reallocation of labor across different labor markets, and finally, mitigating the moral hazard problems arising from unemployment benefits (Calmfors, 1994).

Hiring subsidies, classified as a type of active labor market program, primarily aim to boost labor demand—and thus employment—by temporarily reducing labor costs. These programs help shorten the duration of unemployment by facilitating better matches between workers and jobs. Additionally, they reduce employers' uncertainty about a candidate's employability by substituting formal work experience (Calmfors, 1994; Brown, 2015). With these mechanisms, such programs are seen as tools to reduce labor market inequalities among different population groups.

The effectiveness of hiring subsidies provided to employers for hiring disadvantaged groups has been widely examined. Most of these studies find that the programs increase employment among the target groups, though some have found no significant impact. One reason hiring subsidies may fail to yield the desired impact is the low wage elasticity of labor demand and supply (Katz, 1996; Boockmann et al., 2012). Furthermore, deadweight loss, substitution effects, and displacement effects can limit or even eliminate the employment-generating impact of such programs (Brown, 2015). Deadweight loss occurs when a subsidized individual would have been hired even in the absence of the subsidy. The substitution effect refers to employers preferring certain skill groups over others due to relative changes in hiring costs. The displacement effect arises when subsidized individuals are hired in place of non-subsidized workers.

An example of a hiring subsidy targeting disadvantaged groups is the subsidy program implemented in France during the 1990s for employers hiring low-wage workers. Using household-level

data, Kramarz and Philippon (2001) analyzed how changes in labor costs affected individuals' transition probabilities between employment and unemployment. They compared those directly affected by minimum wage changes with those earning just above the new threshold. Their findings suggest that rising labor costs increased the likelihood of transitioning into unemployment. However, the decrease in labor costs from insurance premium support had no significant effect on transitions from unemployment to employment. The study also mentioned a substitution effect among individuals affected by the premium support but did not quantify its magnitude. Using firm-level data, Crepon and Desplat (2002) found that firms benefiting more from the program experienced greater employment growth compared to others. A key distinction between the French program and the one analyzed in this thesis is that the former lacked a clearly defined target group, making causal identification more difficult.

Studies examining the impact of hiring subsidies with clearly defined target groups generally employ Difference-in-Differences (DiD) and Regression Discontinuity Design (RDD). Some also use Propensity Score Matching (PSM) to improve comparability between treatment and control groups. Blundell et al. (2004) investigated the New Deal program in the UK targeting individuals aged 18–24. Since program eligibility was age-based, they employed DiD combined with PSM. The results showed the program increased the employment probability of young individuals by 5 percentage points, although this effect was not long-lasting. De Giorgi (2005) used the RDD approach to estimate the long-term impact of the same program and found it raised youth employment by approximately 4.6%. Boockmann et al. (2012) examined Germany's EGZ program, which promoted employment of older individuals, using a DiD approach. Their results indicate that the program increased employment only among older women in East Germany, while causing substantial deadweight losses for older men. Huttunen et al. (2013) analyzed the impact of insurance contribution discounts for employers hiring elderly, low-wage workers in Finland using a Triple Difference approach but found no significant effects.

In Turkey, a key example of hiring subsidies targeting disadvantaged groups is the 2008 Employment Package.² The 2008 program provided insurance premium support to employers hiring men aged 18–29 and women of all ages. Many studies have evaluated its impact, typically using the DiD method and focusing on women. Uysal (2013), using macro-level data, found that the prog-

²Other examples of hiring subsidies in Turkey include the regional incentive programs implemented under Law No. 4325 in 1998, amended by Law No. 5084 in 2004 and Law No. 5350 in 2005, as well as the 6-point premium discount introduced in 2013 under Law No. 6486 for certain regions. These programs did not have demographic targeting. Their labor market effects have been analyzed by Betcherman et al. (2010) and Aşık et al. (2022), respectively.

ram increased women's employment in the formal sector. Balkan et al. (2016a), using microdata from the Income and Living Conditions Survey (SILC), reported similar findings. Some studies noted that regional incentive programs might limit the impact of the 2008 program and applied DiD using regional variation. Dildar (2020), for instance, found that women's employment increased in regions without overlapping regional subsidies. Ayhan (2013) used a Triple Difference method to isolate the effect of the 2008 crisis and found that the program increased women's employment shortly after its announcement, but its impact diminished over time.

The studies most similar to this thesis are Balkan et al. (2016b) and Özdamar et al. (2021). Balkan et al. (2016b) estimated the effect of the 2008 program on both young men and women using HLFS data and DiD. Men aged over 30 were used as the control group. The results showed that while women benefited from the program, there was no significant effect for young men. Özdamar et al. (2021), on the other hand, used SILC data and RDD to evaluate the program's effect on young men and found that it did increase their employment likelihood. They also criticized Balkan et al. (2016b) for selecting men as the control group when estimating the effect on women and argued that DiD might suffer from selection bias. They emphasized the reliability of the RDD approach. In contrast, this thesis employs both DiD and RDD methods, utilizing the age-based eligibility rule of the 2018 Additional Employment Program. This allows us to account for methodological differences in results. Additionally, Propensity Score Matching was not deemed necessary for DiD estimation, as the program's impact was re-estimated using variables that clearly differentiate the treatment and control groups.

This study contributes to the literature by analyzing a different active labor market program in the context of the Turkish labor market. Moreover, since the program was implemented during a non-crisis period, it allows for a cleaner estimation of the causal effect of hiring subsidies on employment outcomes.

CHAPTER III: POLICY BACKGROUND

This section examines the structure of the incentive programs that were in effect between 2016 and 2019. It first outlines the legal basis of the 2018 Additional Employment Incentive—whose causal impact is analyzed in this thesis—as well as the 2008 Employment Program and the support provided to those who completed on-the-job training programs, which similarly target the labor market outcomes of the same demographic group. It then discusses other incentive schemes that were not designed based on demographic characteristics.

3.1. ADDITIONAL EMPLOYMENT PROGRAM

The additional employment incentive package, which took its final form in 2018, was initially implemented in February 2017. According to the temporary Articles 17 and 18 of Law No. 4447, employers could benefit from insurance premium and income tax withholding supports for employees hired in addition to their average number of workers until the end of 2017. Under temporary Article 17, the employer could receive support in the amount calculated by multiplying the monthly number of premium payment days of the newly hired employee by 22.22 TL. Under temporary Article 18, an income tax reduction was applied for the amount calculated based on the gross monthly minimum wage in 2017 and the number of premium payment days, after deducting the amount of minimum living allowance. The employer was entitled to benefit from only one of these incentives.

In 2018, with the addition of temporary Article 19 to Law No. 4447 and the issuance of Circular No. 2018/22, the amount of incentive and duration of subsidy provided to employers were changed. For workplaces operating in the manufacturing and IT sectors, the full amount of both the insured and employer shares of premiums calculated on earnings subject to premiums—up to the gross minimum wage—was subsidized. For workplaces in other sectors, the full amount of both the insured and employer shares of premiums was subsidized based on the lower limit of the earnings subject to premiums. The incentive amount based on the lower limit of SPEK (earnings subject to premiums) is 1,876.50 TL for all sectors. This amount also applies as the maximum support in sectors other than manufacturing and IT. However, if the employer operates in the manufacturing or IT sector, they can receive a subsidy of up to 5,004.00 TL based on the upper limit of SPEK.

Within the scope of this subsidy, the employer may benefit from the insurance premium support for 12 months for each worker employed in addition to the average number of workers. If the newly employed individual is a male aged between 18 and 25 or a woman over 18, this duration is extended to 18 months.

3.2. 2008 EMPLOYMENT PROGRAM

Another program implemented concurrently with the 2018 Additional Employment Program examined in this thesis is the 2008 employment package. This program entered into force in July 2008 and provided support for the employer's share of insurance premiums for a period of five years for employers who employ men over the age of 18 and under the age of 29, as well as women. According to the program, it was decided that the employer's share of insurance premiums would be covered by the Unemployment Insurance Fund at the rates of: 100% for the first year, 80% for the second year, 60% for the third year, 40% for the fourth year, and 20% for the fifth year. With Law No. 5838, the duration of this program was extended until the end of 2010.

In 2011, with Law No. 6111, the structure and scope of the incentives changed. The gradual reduction of the employer's share of insurance premiums on a yearly basis was abandoned, and it was decided that the full amount would be subject to a discount for each year the incentive is applied. The duration of the incentive varies depending on the characteristics of the employed individual. Table 1 presents the structure of the employment package. The program is still ongoing and is planned to remain in effect until the end of 2022. The name of the subsidy package is *Insurance Premium Support for the Employment of Youth, Women, and Those with Vocational Certificates*.

Table 1. 2008 Employment Program

Voc. Certificate	İŞKUR	Voc. Training	Incentive Duration
Males and Females Aged 18–29			
√	-	-	48 Months
√	√	-	54 Months
-	-	√	36 Months
-	√	√	42 Months
-	-	-	30 Months
-	-	-	24 Months
Males Over the Age of 29			
√	-	-	24 Months
-	-	√	24 Months
√	√	-	30 Months
-	√	√	30 Months
-	-	-	6 Months

3.3. TRAINING COMPLETION INCENTIVE PROGRAM

According to the temporary Article 15 of Law No. 4447, employers who employ individuals over the age of 18 and under the age of 29 within three months of completing on-the-job training programs organized by İŞKUR are provided with employer share support for insurance premiums for a period ranging from 30 to 48 months. The program, which was implemented on April 23, 2015, is still in effect.

3.4. OTHER EMPLOYMENT PROGRAMS

In Turkey, many employment incentive programs were implemented between 2016 and 2020. These can be listed, as follow. These incentive programs do not have any demographic characteristics.

3.4.1. 5-Point Premium Discount

According to Article 81, first paragraph (ı) of Law No. 5510, a 5-point discount is applied on the employer's share of disability, old-age, and death insurance premiums calculated based on the insured persons' earnings subject to premium payments, which are paid by private sector employers. This support has been in effect since 01/10/2008 and continues to be applied today.

3.4.2. 6-Point Premium Discount

With Law No. 6486, a paragraph was added to Article 81 of the Social Insurances and General Health Insurance Law No. 5510, providing additional premium discounts of up to 6 points based on the lower limit of the earnings subject to premiums, in addition to the 5-point social security premium support, for private sector employers operating in 51 provinces with low socio-economic development index and in the districts of Bozcaada and Gökçeada. This support was in effect from January 1, 2013, until the end of 2021.

3.4.3. Investment and Employment-Based Premium Support

According to the Additional Article 2 of Law No. 5510 and Communiqués No. 2009/1 and 2012/1, within the framework of decisions on state aid in investments and incentive certificates, for insured persons employed through investments, the social security premiums calculated based on the earnings subject to premiums determined under Articles 81 and 82, including the employer's share entirely or together with the insured's share up to the limit determined by the President considering statistical region classification, per capita national income, or socio-economic development levels, are covered from the Ministry of Economy budget. This incentive has been in effect since 2009.

3.4.4. Incentive for Employing Those Receiving Unemployment Benefits

According to Article 50 of Law No. 4447 and General Directive No. 2009/149, if an insured person who is entitled to unemployment benefits is employed by the employer in addition to the average number of workers declared in the last six months, the short-term insurance premiums calculated over the earnings subject to premiums for the insured's share, employer's share, and general health insurance premium, amounting to 1% of the insured's share, are covered as a discount for the remaining unemployment benefit period. This incentive package has been in effect since October 1, 2009.

3.4.5. Employment of Disabled Persons

According to Article 30 of Law No. 4857 and General Directive No. 2008/77, the entire employer's share of the social security premium calculated over the earnings subject to premiums for disabled insured persons employed in private sector workplaces will be covered.

3.4.6. R&D Design Incentive

According to Temporary Article 2 of Law No. 4691, Article 3 of Law No. 5746, and General Directives Nos. 2008/85 and 2009/21, half of the employer's share of the social security premium calculated on wages of R&D design and support personnel whose salaries are exempt from income tax will be covered by the Ministry of Finance until the end of 2023.

3.4.7. Culture Investment/Entrepreneurship Incentive

According to Article 5 of Law No. 5225 and General Directive No. 2010-109, for insured persons working in workplaces that have obtained a Culture Investment Certificate, 50% of the employer's share of the social security premium calculated on earnings subject to premiums will be covered for 3 years, and for insured persons working in workplaces with a Culture Entrepreneurship Certificate, 25% of the employer's share of the social security premium will be covered for 7 years.

3.4.8. Incentive for Children Benefiting from Social Services

According to Additional Article 1 of Law No. 2828 and General Directive No. 2018/7, since 19.02.2014, for the employment of children benefiting from social services, the entire insured and employer share of the social security premium and unemployment insurance premium calculated on the earnings subject to premium limits will be provided as a discount for five years for each insured.

3.4.9. Incentive for Those Receiving Social Assistance

According to Additional Article 5 of Law No. 3294 and General Directive No. 2018/8, for the employment of those receiving social assistance, the entire employer's share of the social security premium calculated on the earnings subject to premiums will be covered by the Ministry of Labor and Social Services.

CHAPTER IV. THEORETICAL BACKGROUND

Active labor market policies are among the foremost policies adopted by governments both in combating unemployment and informal employment, and in addressing inequalities in the labor market. However, the findings obtained from studies on the effectiveness of these policies do not lead us to a clear conclusion. Although many studies in the literature conclude that these policies increase formal employment, there are also studies that claim the opposite. Because the additional effects that emerge alongside these policies may prevent the expected impact of the policies from materializing (Calmfors, 1994). On the other hand, these policies also differ in terms of implementation, and their effects on the labor market show heterogeneity (Calmfors, 1994; Calmfors and Skedinger, 1995; Heckman et al., 1999).

Heckman et al. (1999) examine active labor market policies in five different groups: (a) classroom training, (b) subsidized employment in the public and private sectors, (c) on-the-job training supports, (d) training programs for vocational acquisition, (e) support applications in job search activities. Calmfors (1994) states that these policy versions will affect the labor force through different channels. The 2018 Additional Employment Program is evaluated under group (b) due to the social security discount it provides to employers for each additional employee, and in this section, the theoretical framework of policy implementations in this group is examined. While explaining the theoretical framework, Calmfors (1994) is taken as the basis and the effects of these policies on the labor market are explained through the Layard Nickell Model and the Beveridge Curve. Then, the adverse effects that reduce the effectiveness of these policies are discussed.

4.1. LAYARD NICKELL MODEL

Calmfors (1994), while theoretically explaining active labor market policies, uses the model developed by Richard Layard and Stephan Nickell (See: Layard and Nickell, 1986; Johnson and Layard, 1987; Layard et al., 1991). In Figure 1, a graphical representation of the model is presented. The downward sloping curve in the graph is called the *employment schedule* and is assumed to be equal to labor demand. The upward sloping curve is called the *wage-setting schedule*. This curve shows how high employment rates create pressure for an increase in real wages. Calmfors (1994) explains this by giving the example of employer unions. Accordingly, during periods when employment rates are high and job vacancies are abundant, union activities aiming to reduce wages decline. Moreover, in such a conjuncture, the employer increases the real wage amount to enhance competitiveness for labor and to extract optimal effort from employees. The intersection of these two curves indicates the equilibrium in the labor market. In Figure 1, the equilibrium is represented by point A. The distance between the full employment and the equilibrium point is called the *involuntary unemployment rate* (u_0).

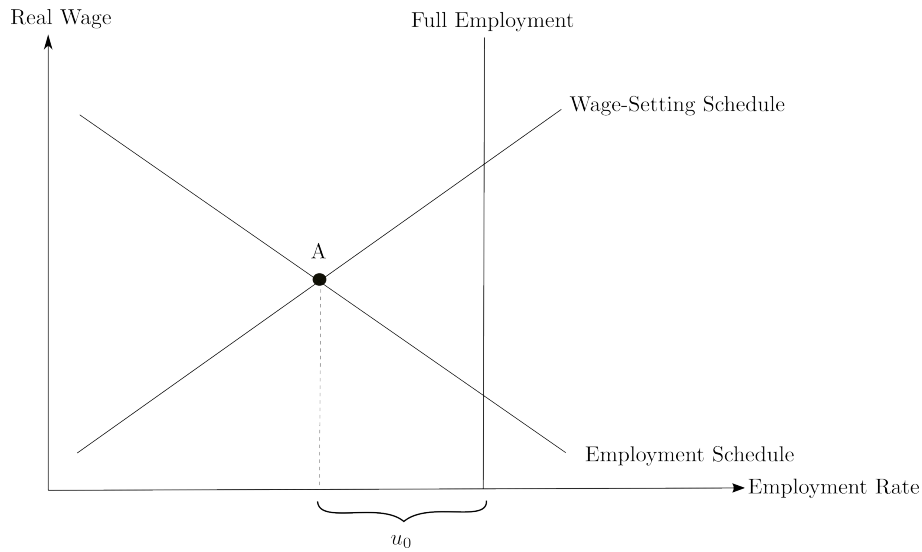


Figure 1. Layard Nickell Model (1986)

Drawing attention to the *matching process* between employers and workers in active labor market policies, Calmfors (1994) states that subsidized employment-themed policy implementations function as a *screening tool* for employers. Accordingly, active labor market policies aiming to create employment by reducing labor costs serve as a discount factor (Brown, 2015; Özdamar et al., 2021). Because there is uncertainty for the employer when hiring an unqualified individual, due to the lack of information about their marginal productivity. These policies replace work experience and reduce the uncertainty between employer and worker, thereby encouraging employers to

demand more labor. This situation is shown in Figure 2 by the rightward shift of the employment program curve. The new equilibrium is point A' . At this point, employment will increase.

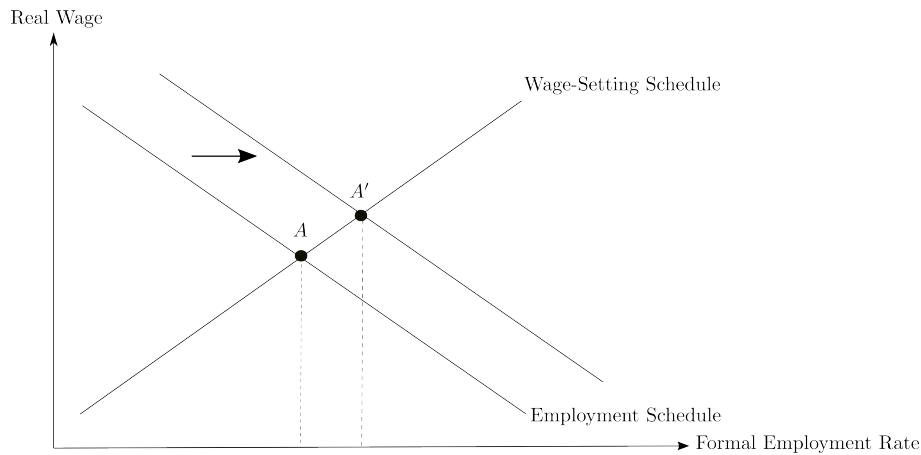


Figure 2. The Effect of Active Labor Market Programs in the Formal Sector

Over time, as employment increases and job vacancies decrease, employers will refrain from offering higher real wages to retain their labor force. This will cause the wage-setting curve to move downward. As a result of the movement of both the employment and wage-setting curves, the involuntary unemployment rate will fall. Calmfors (1994) states that in the new equilibrium, employment will increase, but the direction of the change in real wages will be uncertain. This new equilibrium situation applies to the formal sector where the program is implemented. The effect of active labor market policies in the informal sector is different. Since the cost of formal employees for the employer becomes relatively lower compared to informal employees, the employer will reduce the amount of labor demanded in the informal sector. This causes the labor demand curve to shift leftward, thereby leading to a decrease in employment rates in the informal sector. Figure 3 shows the effect of active labor market policies implemented in the formal sector on the informal sector.

4.2. BEVERIDGE CURVE

Calmfors (1994), while theoretically explaining active labor market policies, also uses the Beveridge Curve in addition to the Layard Nickell Model. The Beveridge Curve refers to the negative relationship between the *vacancy rate* and the unemployment rate and is a tool that provides information about the matching efficiency between job vacancies and unemployed individuals in the labor market. Movements along the Beveridge Curve are related to cyclical shocks. During recession periods, the vacancy rate decreases while the unemployment rate increases. Therefore, the equilibrium in the labor market will form at the upper part of the curve. During economic expan-

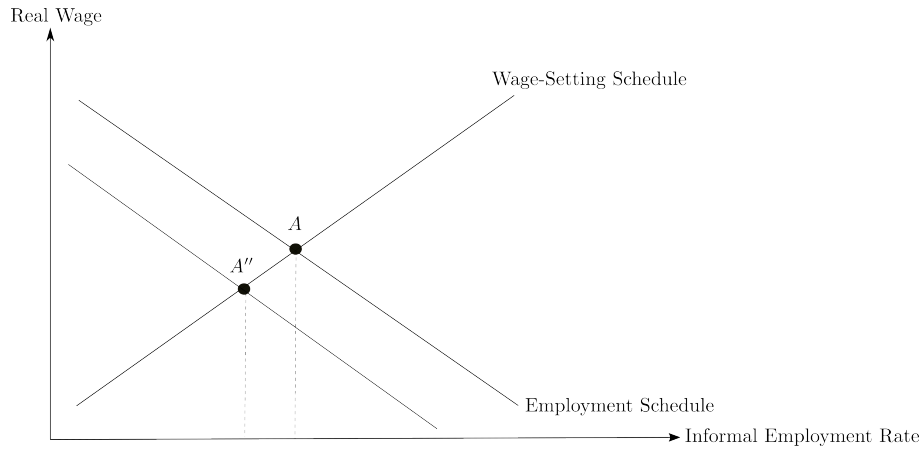


Figure 3. The Effect of Active Labor Market Programs in the Informal Sector

sion periods, the vacancy rate increases while the unemployment rate decreases. In this case, the equilibrium point will be at the lower part of the curve. Changes in the position of the curve are related to structural shocks that affect the matching between vacancies and unemployment (Bouvet, 2012). In Figure 4, the Beveridge Curve is shown, and the unemployment rate at the equilibrium point indicated as A should be the same as the unemployment rate implied by the equilibrium at point A in Figure 1.

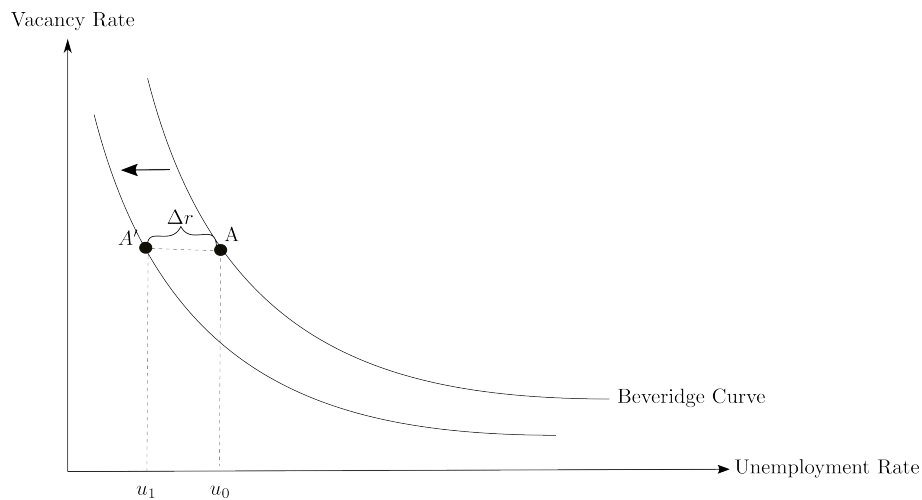


Figure 4. Beveridge Curve

Since hiring subsidies contribute to reducing the uncertainty environment between employers and the unemployed, the matching process between vacancies and the unemployed in the labor market will improve. In this case, the Beveridge Curve will shift toward the origin as shown in Figure 4. At the new equilibrium point, the unemployment rate will decrease. However, the change in the position of the Beveridge Curve applies to the formal sector. Because subsidies are applied to employers operating in the formal sector.

4.3. ADVERSE EFFECTS

When evaluating the effectiveness of hiring subsidies, *deadweight costs*, *substitution effects*, and *displacement effects* should also be considered. Deadweight cost refers to the share of subsidies that goes to workers who would have been employed even in the absence of the subsidy. The substitution effect occurs when firms prefer a particular skill group to perform the same job due to relative changes in labor costs between groups with different qualifications. For example, subsidies developed for the employment of low-skilled individuals may cause employers to prefer low-skilled workers over medium-skilled ones. Finally, hiring subsidies may encourage firms to hire subsidized individuals instead of non-subsidized ones or to lay off existing employees to hire subsidized workers. In fact, when the employment support ends, it is also possible for firms to lay off subsidized workers. This situation is referred to as the displacement effect (Brown, 2015). These effects cause the employment schedule in the formal sector to shift leftward and reduce the employment rate. Calmfors (1994) noted that deadweight and substitution effects are more likely to arise in programs subsidizing young people and the long-term unemployed in the private sector.

CHAPTER V. DATA AND DESCRIPTIVE STATISTICS

In this thesis, the dataset used is the HLFS (Household Labour Force Survey) data regularly published each year by the Turkish Statistical Institute (TurkStat). The HLFS dataset contains detailed information about the economically active population: such as economic activity, occupation, employment status, working hours, and social security registration for the employed; and job search duration and desired occupation and similar characteristics for the unemployed. Additionally, the HLFS provides information about the demographic characteristics of the population such as age, gender, education, marital status, and region of residence. Settlement information in the HLFS is available at the NUTS1 (12 regions) and NUTS2 (26 regions) levels. The HLFS covers all private households living within the borders of the Republic of Turkey, excluding schools, dormitories, day-care centers, nursing homes, private hospitals, military barracks, and civil servant lodgings. In this form, the HLFS constitutes the main data source on the Turkish labor market. Although the HLFS is a cross-sectional micro-level dataset, macro-level information on the Turkish labor market can also be obtained since weights are assigned to each individual in the survey. In the econometric analyses conducted within the scope of the thesis, weights were not used; however, they were employed in the construction of descriptive statistics.

To examine the effects of the 2018 Additional Employment Program on the labor market, observations from the HLFS between the years 2016 and 2019 were used. The years 2016 and 2017 were considered the pre-program period, while 2018 and 2019 were considered the post-program period. In this form, the dataset contains 1,500,123 observations. For the econometric analyses conducted in the study, the dataset was restricted to include only males aged 18–28, and the sample size is 125,651. Furthermore, agricultural workers were excluded from the dataset due to seasonal changes in employment status.

Descriptive statistics are shown in Table 2. For males aged 18–24, the proportion of individuals who have not completed any school and those who graduated from primary and lower-secondary education has decreased over time. There is an increase in the proportion of high school and university graduates. For males aged 25–28, there is no observable trend in the proportion of individuals who have not completed any school or those who graduated from primary and lower-secondary

education; however, the proportion of high school and university graduates has increased over time. The difference in the proportion of university graduates between the two groups can be explained by the fact that individuals aged 18–24 are more likely to still be in education. In both groups, marriage rates have decreased over the years, but the average marriage rate for men aged 25–28 is about five times higher than that of those aged 18–24.

Table 2. Descriptive Statistics

	2016	2017	2018	2019
<i>Treatment Group: Male aged 18-24</i>				
No School	0.052	0.047	0.046	0.043
Primary/ Secondary School	0.421	0.402	0.394	0.373
High School	0.401	0.425	0.432	0.451
University	0.127	0.125	0.128	0.132
Married	0.084	0.080	0.074	0.068
Formal Employment	0.343	0.337	0.358	0.317
Informal Employment	0.126	0.127	0.116	0.119
Formal Real Wage	91.127	69.535	46.965	100.298
Informal Real Wage	60.686	50.293	33.181	68.323
Unemployment Rate	0.130	0.136	0.134	0.175
<i>Control Group: Male Aged 25-28</i>				
No School	0.054	0.047	0.050	0.054
Primary/ Secondary School	0.330	0.349	0.335	0.333
High School	0.282	0.258	0.258	0.249
University	0.334	0.346	0.357	0.364
Married	0.387	0.379	0.378	0.354
Formal Employment	0.606	0.606	0.598	0.565
Informal Employment	0.083	0.084	0.083	0.083
Formal Real Wage	118.934	97.603	65.033	135.997
Informal Real Wage	69.458	59.209	38.730	87.460
Unemployment Rate	0.124	0.124	0.126	0.159
<i>Observation: Treatment Group: 14,979,434 & Control Group: 9,187,812</i>				

Looking at labor market indicators, the formal employment rate for males aged 25–28 is around 60%, which is higher compared to males aged 18–24. The group with the highest rate of informal employment is young males aged 18–24. On the other hand, hourly real wages are higher for males aged 25–28 compared to younger males in both the formal and informal sectors. The fact that informal employment rates are higher among young individuals while their real wages are lower compared to adults is an indicator of inequality in the labor market. The variation in real wages across years in both formal and informal sectors can be explained by the instability in inflation

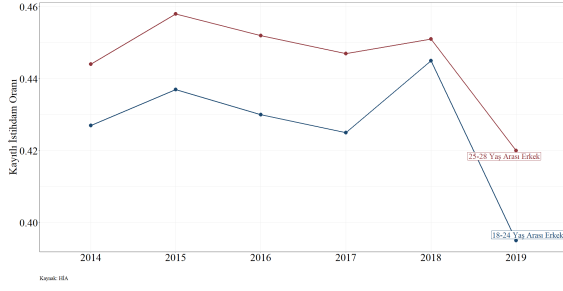
rates. The annual inflation rates for 2016, 2017, 2018, and 2019 were 8.5%, 11.92%, 20.3%, and 11.84%, respectively. In terms of unemployment rates, unemployment is higher for males aged 18–24. In 2019, the unemployment rate increased by approximately 4 percentage points compared to the previous year.

Table 3 shows how the mean values of variables for the treatment and control groups changed before and after the subsidy package. The average difference between the two groups changed very slightly for the explanatory variables used in the model. This is important in demonstrating that the composition of the control and treatment groups did not change significantly after the program.

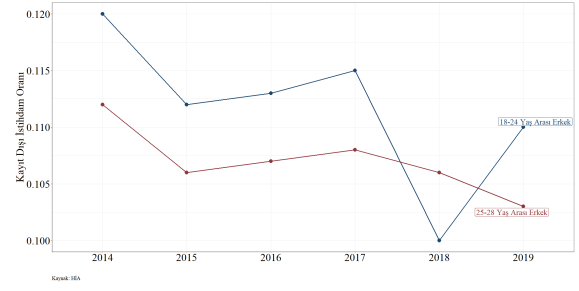
Table 3. Descriptive Statistics for Pre- and Post-Program

	Treatment Group (18-24)		Control Group (25-28)		Difference (Treatment-Control)	
	2018 ⁻	2018 ⁺	2018 ⁻	2018 ⁺	2018 ⁻	2018 ⁺
No School	0.050	0.044	0.051	0.052	-0.001	-0.007
Primary/ Secondary School	0.412	0.384	0.340	0.334	0.072	0.050
High School	0.413	0.442	0.270	0.254	0.143	0.188
University	0.126	0.130	0.340	0.361	-0.214	-0.230
Married	0.082	0.071	0.383	0.366	-0.301	-0.295
<i>Observation: Treatment Group: 14,979,434 & Control Group: 9,187,812</i>						

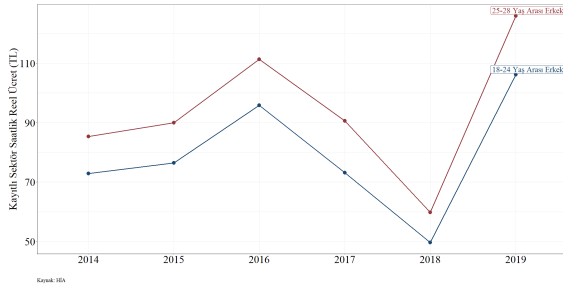
Graphs showing the employment rates and real wages in the formal and informal sectors, as well as the unemployment rates of the treatment and control groups by year, are presented in Figure 5. Accordingly, there is a similar time trend between the treatment and control groups, particularly in formal sector employment, real wages, and unemployment rates. However, in the informal sector, different trends appear after 2017. Nevertheless, this does not apply to the pre-treatment period.



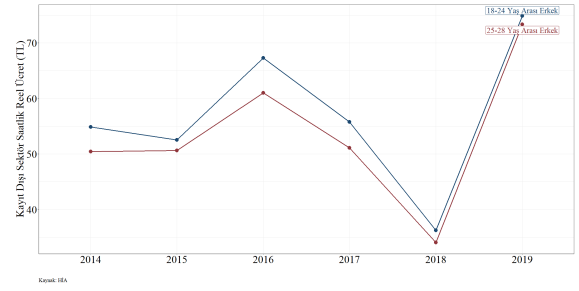
Formal Employment Rate



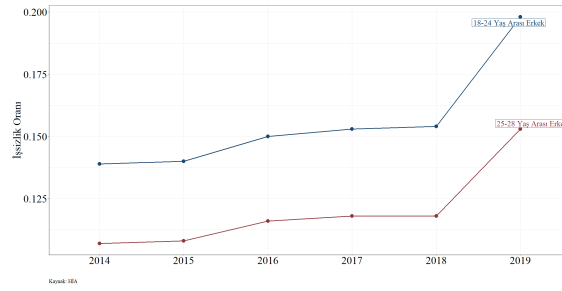
Informal Employment Rate



Formal Real Wage



Informal Real Wage



Unemployment Rate

Figure 5. Labor Market Outcomes by Years

CHAPTER VI. ECONOMETRIC STRATEGY

In this thesis, the Difference-in-Differences method and the Regression Discontinuity Design are used to measure the impact of the 2018 Additional Employment Program on the labor market outcomes of young men.

6.1. DIFFERENCE-IN-DIFFERENCES METHOD

The 2018 Additional Employment Program targeting the employment of young men aged 18-24 allows us to analyze the causal effect of this program using the Difference-in-Differences method. In this section, first, how the causal effect of the employment program is analyzed with the Difference-in-Differences method is explained, then the identification strategy and the estimated econometric model are introduced. Additionally, the *parallel trends assumption* required for the application of the Difference-in-Differences method is tested.

6.1.1. Design

The fundamental problem of causal inference is that it is impossible to observe a person's potential outcomes in two different states at the same time. However, we can overcome this problem by forming new groups to represent the counterfactual state that we cannot observe. At this point, the Difference-in-Differences method is one of the main research designs used to estimate causal effects in cases where treatment assignment is not random.

As explained in the policy framework section, age is the sole determinant of whether a person is affected by the program. Since no regional differentiation was considered during program implementation, it is not possible to reach the causal effect by comparing the outcomes of individuals in the same age group before and after the intervention. The effect of programs without regional breakdowns but with age-dependent treatment can be measured by choosing individuals of different ages as the control group. The Additional Employment Program, whose causal effect is attempted to be estimated in the thesis, targets the employment of young men aged 18-24. Therefore, by choosing men over 24 as the control group, it is possible to estimate the causal effect using the Difference-in-Differences method. However, the most important point to consider here is that the treatment and control groups should have similar trends over time.

6.1.2. Identification Strategy

Let Y_{it}^1 denote the labor market outcome of individual i in period t when affected by the Additional Employment Program. Let Y_{it}^0 denote the labor market outcome of the same individual i in period t when not affected by the program, i.e., the counterfactual outcome. The causal effect for individual i can be expressed as $Y_{it}^1 - Y_{it}^0$. The average treatment effect of the program, denoting individuals affected by the program as $D = 1$, can be expressed as $E[Y_{it}^1 - Y_{it}^0 | D = 1]$. However, since we only observe one outcome for a person in period t , we cannot calculate the causal effect in this way. Here, we can reveal the causal effect by choosing an alternative group for $E[Y_{it}^0 | D = 1]$.

Based on the age cutoff design of the program, let's take men aged 25-28 as the control group. In this case, for men younger than 25, $D = 1$, and for older men, $D = 0$. Let the pre-2018 period be $t = 0$, and the post-period be $t = 1$. If we assume that in the absence of treatment the labor market outcomes of these two groups follow similar changes over time, the unobservable counterfactual outcome can be written as in Equation (1). In this equation, τ represents the difference in trends between the two groups.

$$E[Y_{it}^0 | D = 0, t = 1] = E[Y_{it}^0 | D = 0, t = 0] + \tau \quad (1)$$

The difference of the treated group's outcomes before and after treatment can be expressed as in Equation (2). Here, γ represents the causal effect of the program.

$$E[Y_{it}^1 | D = 1, t = 1] - E[Y_{it}^1 | D = 1, t = 0] = \tau + \gamma \quad (2)$$

Note that the difference in labor market outcomes for the treated group before and after treatment consists of trend and causal effect components. If the treatment and control groups move similarly over time, i.e., share a common trend, we can subtract Equation (1) from Equation (2) to obtain the causal effect. This is the basic principle of the Difference-in-Differences method.

$$\begin{aligned} & E[Y_{it}^1 | D = 1, t = 1] - E[Y_{it}^1 | D = 1, t = 0] - \\ & (E[Y_{it}^0 | D = 0, t = 1] - E[Y_{it}^0 | D = 0, t = 0]) = \gamma \end{aligned} \quad (3)$$

6.1.3. Model

Equation (4) presents the model developed for estimating the Difference-in-Differences method. Indices i , r , and t denote individual, region, and year, respectively. y_{irt} is a dummy variable indicating whether the person is employed in the formal or informal sector. When the employment variable is the dependent variable, estimation is done using the *linear probability model*. The logarithm of hourly real wages in the formal and informal sectors is also used as a dependent variable. When measuring the effect of the employment program on real wages, a *linear regression model* is employed.

$TargetGroup_i$ is a dummy variable equal to 1 for males aged 18-24 and 0 for males aged 25-28. $Post18_t$ takes the value 1 for years 2018 and 2019, and 0 for 2016 and 2017. X_i is a vector of control variables expressing the individual's characteristics and their interaction terms. Education level, marital status, and age are used as control variables in the model. α_t and α_r denote year and region fixed effects, respectively. $Trend_r$ shows region-specific linear trends. Thus, regional factors changing over time are controlled for. An example of such factors includes regionally implemented subsidy programs. Although regional subsidies are not demographic in nature, they can affect the labor market outcomes of the target group and therefore are controlled for. ϵ_{it} denotes the error term, clustered at the NUTS2 region level. The program effect is represented by the coefficient δ_{DD} of the interaction term between $TargetGroup_i$ and $Post18_t$.

$$y_{irt} = \beta_0 + \beta_1[TargetGroup_i] + \beta_2[Post18_t] + \delta_{DD}[TargetGroup_i * Post18_t] + \beta_3X_i + \alpha_t + \alpha_r + Trend_r + \epsilon_{it} \quad (4)$$

To show how the program effect changes by year, Equation (5) is estimated. In the model, the coefficient δ_{DD}^n represents the causal effect of the program for each post-intervention year.

$$y_{irt} = \beta_0 + \beta_1[TargetGroup_i] + \beta_2[Post18_t] + \delta_{DD}^n \left[TargetGroup_i * \sum_{n=2018}^{2019} (Year = n) \right] + \beta_3X_i + \alpha_t + \alpha_r + Trend_r + \epsilon_{it} \quad (5)$$

6.1.4. Parallel Trends Assumption

The Difference-in-Differences method relies on the assumption that the treatment and control groups would move similarly over time in the absence of any policy effect. This assumption, called Pa-

parallel Trends, if violated, results in biased estimates. Because the potential outcomes of the treated group after intervention are unobservable, the assumption can only be tested in the pre-intervention period.

The model developed to check the parallel trends assumption is shown in Equation (6). Indices i , r , and t denote individual, region, and year, respectively. y_{irt} denotes dummy variables indicating employment status in the formal and informal sectors and the logarithmic hourly real wages in both sectors. The variable $TargetGroup_i$ takes the value 1 for individuals aged 18-24, and 0 for those aged 25-28. α_t represents year fixed effects. Error terms, clustered at the NUTS2 regional level, are denoted by ϵ_{it} .

$$y_{irt} = \sum_{n \neq 2017} \beta_s [TargetGroup_i * (Year = n)] + \alpha_t + \epsilon_{it} \quad (6)$$

Table 4. Parallel Trend Test

	Formal		Informal	
	(1) Employment	(2) Real Wage	(3) Employment	(4) Real Wage
2015	-0.012 (0.010)	-0.012 (0.012)	0.00052 (0.005)	0.032 (0.037)
2016	-0.0013 (0.009)	0.013 (0.016)	-0.0010 (0.005)	0.016 (0.036)
2018	0.021** (0.009)	-0.019 (0.014)	-0.013*** (0.005)	0.030 (0.031)
2019	0.016 (0.010)	-0.015 (0.022)	-0.011** (0.005)	-0.026 (0.039)
<i>Observation</i>	159,023	60,495	159,023	13,100

Note: The model uses observations from the HLFS dataset between 2015 and 2019. The target group is defined as young men aged 18-24. The control group is comprised of male individuals aged 25-28. Estimates for the employment variable are obtained using a linear probability model. The wage variable is generated by taking the logarithm of the hourly real wage reported by individuals. The model includes dummy variables for education, age, marital status, and their interactions. Robust standard errors are clustered according to the NUTS2 regional classification and are presented in parentheses. ***, **, and * indicate 1%, 5%, and 10% significance levels, respectively.

In Equation (6), the year 2017, the period just before program implementation, is chosen as the reference year. Since there is no intervention effect for males aged 18-24 prior to the program's enactment, the coefficients β_s in Equation (6) represent the trend difference between the treatment and control groups. The estimates obtained from this model are presented in Table 4. In the pre-intervention period, the trend difference between the treatment and control groups for the four labor market outcomes considered in the thesis is statistically insignificant compared to 2017. Therefore,

these results show that the parallel trends assumption holds in the pre-intervention period.

The graphical presentation of the parallel trends assumption test results is shown in Figure 6. Figure 6 displays the trend differences between the treatment and control groups for the four main labor market outcome variables as point estimates with 95% confidence intervals for each year. According to Figure 6, for all labor market outcomes, the trend difference between the two groups in the pre-intervention period is not statistically different from zero, indicating that the two groups share a common trend prior to intervention. Therefore, individuals aged 25-28 serve as a good control group in the post-intervention period.

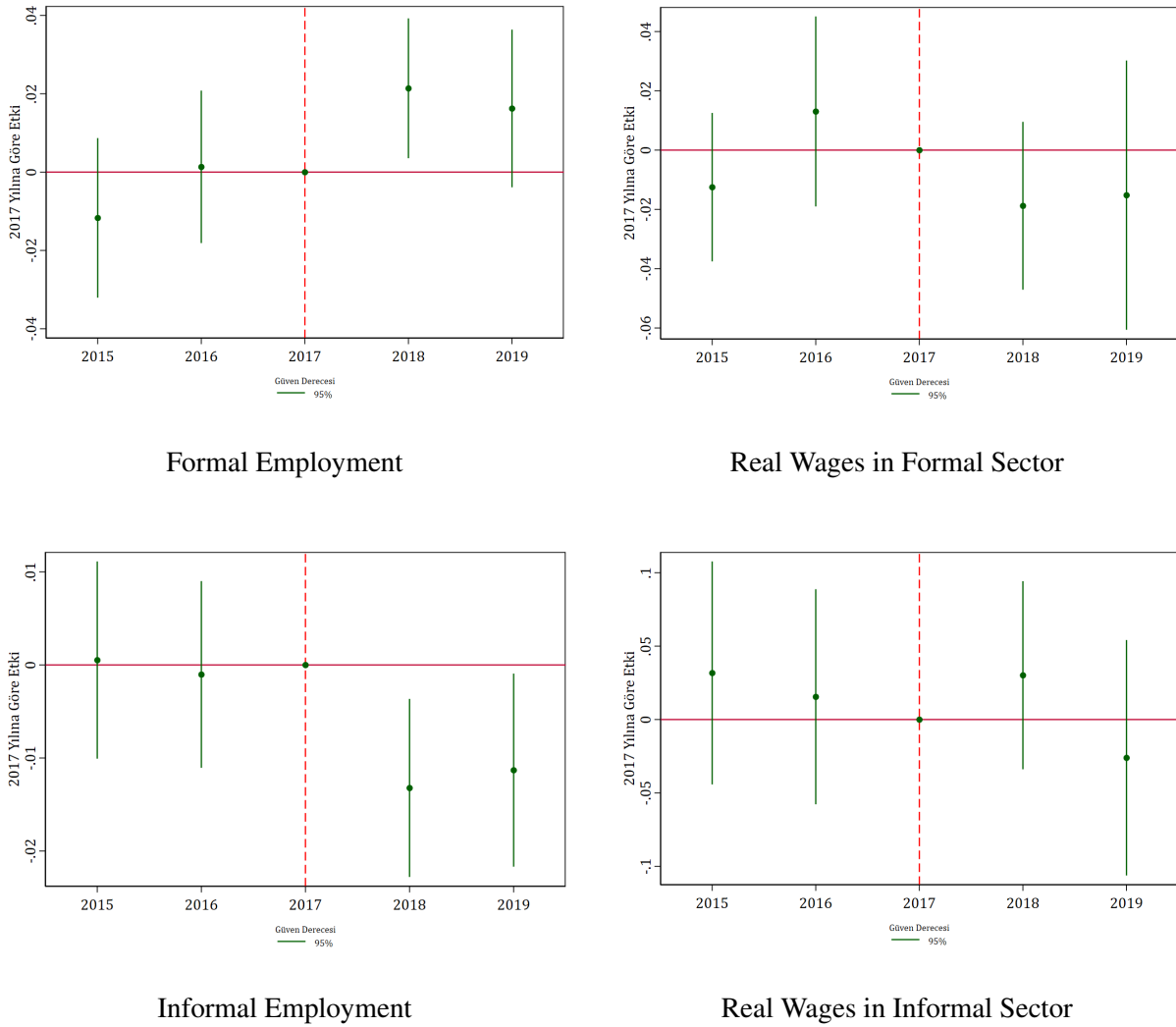


Figure 6. Parallel Trends Assumption

6.2. REGRESSION DISCONTINUITY DESIGN

The Additional Employment Program targeting males under the age of 25 allows us to estimate the causal effect of this program not only with the Difference-in-Differences method but also using the Regression Discontinuity Design. In this section, first, how the program is estimated using the Regression Discontinuity Design is explained, followed by the identification strategy and the estimated econometric model.

6.2.1. Design

The fundamental problem when trying to determine the causal effect of an intervention is that it is impossible to observe a person's outcomes under two different conditions at the same time. When selecting alternative control groups for the unobservable counterfactual situation, we can use other research designs alongside the Difference-in-Differences method. At this point, we may examine the criteria used for the implementation of the intervention.

The Regression Discontinuity Design is constructed based on rules that enable the implementation of any intervention (Angrist and Pischke, 2009). These rules create a randomness in the application of the intervention, and the presence of such rules is considered a natural experiment. If the assignment rule for an intervention is deterministic, that is, the same for all individuals, the design is called a *Sharp Regression Discontinuity Design*, while if the assignment rule is probabilistic, the design is called a *Fuzzy Regression Discontinuity Design* (Cattaneo et al., 2019).

The implementation of the Additional Employment Program for males aged 18-24 without any characteristic or regional differentiation allows us to estimate the effect of this program using the Sharp Regression Discontinuity Design. The participation condition for this program is a deterministic function of age. Therefore, we can estimate the effect of the program by comparing the labor market outcomes of men younger and older than 25. There will be no difference other than the program between those who are just below and just above the cutoff.

In studies aiming to reveal the causal effect using Regression Discontinuity Design, the choice of bandwidth is an important issue. Because narrower bandwidths minimize bias but increase standard errors due to smaller sample size, a bias-variance tradeoff exists, and tests developed to find the optimal bandwidth are widely used. In this study, estimations were made for all possible bandwidths to examine the sensitivity of the results.

6.2.2. Identification Strategy

Let Y_{it}^1 be the potential outcome of individual i in period t if affected by the program, and Y_{it}^0 be the potential outcome if not affected. Since we observe only one outcome for individual i in period t , we need to find a control group. Let D be a dummy variable indicating whether the person received the intervention, the individual's age be a , and the cutoff be c . The intervention assignment is a deterministic function of the individual's age, which can be expressed as in Equation (7):

$$D_{it} = \begin{cases} 1, & a \geq c \\ 0, & a < c \end{cases} \quad (7)$$

If individual i is younger than 25, they benefit from the program, so the cutoff c is defined as 25. The assignment condition can be expressed using the indicator function $\mathbb{1}(\cdot)$ as $D_{it} = \mathbb{1}(a \geq c)$. Let Y_t^+ and Y_t^- be defined as males marginally just above and just below the cutoff age 25, respectively. This can be expressed as in Equation (8):

$$\begin{aligned} Y_t^+ &\equiv \lim_{a \rightarrow c^+} E[Y_{it}|a = c] = E[Y_t|c^+] \\ Y_t^- &\equiv \lim_{a \rightarrow c^-} E[Y_{it}|a = c] = E[Y_t|c^-] \end{aligned} \quad (8)$$

Under the assumption that potential outcomes are continuous at the cutoff and local randomization holds, the Sharp Regression Discontinuity Design identifies the causal effect of the subsidy program by taking the difference between the values just above and just below the cutoff at time t . In Equation (9), γ represents the causal effect of the program:

$$Y_t^+ - Y_t^- = \gamma_t \quad (9)$$

6.2.3. Model

Equations (10) and (11) are used for estimating the Regression Discontinuity model. Equation (10) is developed for a linear estimation of the Regression Discontinuity method, and Equation (11) for a quadratic estimation. Indices i , r , and t denote individual, region, and year, respectively. y_{irt} indicates dummy variables for whether the individual is employed in the formal or informal sector and the logarithmic hourly real wages in both sectors. The variable $TargetGroup_i$ takes the value 1 for individuals aged 23-24 and 0 for individuals aged 25-26. Error terms are clustered at

the NUTS2 region level and denoted by ϵ_{it} . The age of the individual is a , and c shows the age cutoff from the program, set at 25. The causal effect of the program is indicated by the coefficient δ_{RD} .

$$y_{irt} = \beta_0 + \delta_{RD}[TargetGroup_i] + \beta_1[f(age_i - c, a)] + \beta_2[TargetGroup_i * f(age_i - c, a)] + \alpha_r + \epsilon_{irt} \quad (10)$$

$$y_{irt} = \beta_0 + \delta_{RD}[TargetGroup_i] + \beta_1[f(age_i - c, a)] + \beta_2[f(age_i - c, a)]^2 + \beta_3[TargetGroup_i * f(age_i - c, a)] + \beta_4[TargetGroup_i * f(age_i - c, a)^2] + \alpha_r + \epsilon_{irt} \quad (11)$$

CHAPTER VII. ESTIMATION RESULTS

This section presents the estimates that demonstrate the impact of the 2018 Additional Employment Program on the labor market outcomes of young men. Since the program was designed to increase the employment of wage earners in the formal sector, the analysis of the program's overall effect is based on individuals' status as wage earners, and the results are examined in two stages according to whether employees work in the formal or informal sector. In addition, the program's effect on real wages is also presented. The estimates were obtained using the Difference-in-Differences (DiD) method and the Regression Discontinuity Design (RDD), and the findings from each model are presented separately.

7.1. DIFFERENCE-IN-DIFFERENCES METHOD ESTIMATION RESULTS

This section first presents the findings on the impact of the program on the overall labor market outcomes of young men. Then, it provides estimates showing how the program's effect has changed over the years. Next, the program's impact is examined based on the education levels of young men across four groups: those with no formal education, primary or middle school graduates, high school graduates, and finally university (undergraduate and graduate) degree holders. Lastly, the effect of the subsidy package is analyzed by age groups.

This section also includes the results of tests conducted to assess the reliability of the estimates. To evaluate the accuracy of the estimates, a placebo intervention effect for the year 2017 is first presented. The results of the placebo test are reported separately for registered and unregistered wage workers, as well as for real wages in both sectors. Following this, the program's effect is re-estimated using narrower observation windows.

7.1.1. The Effect of Program on Young Male's Labor Outcomes

The impact of the program on youth employment and real wages in the formal sector is presented in Table 5. In the first and fourth columns, the model is estimated for employment and real wage outcomes, respectively, including only control variables. In the second and fifth columns, year and region fixed effects are added to the model. The third and sixth columns report the results of the model specified in Equation (4).

According to these results, the program increased the probability of young men being employed in the formal sector by 2 percentage points, while it reduced their real wages by 2%. For the employment outcome, the effects of fixed effects and regional trends are not particularly significant, as the magnitude and statistical significance of the estimates did not change considerably. However, for the real wage outcome, the fixed effects matter. On the other hand, similar to the employment variable, the regional trend did not create substantial changes in the estimates for real wages either.

Table 5. The Effect of Program on Formal Employment

	Formal Sector					
	Employment			Real Wage		
	(1)	(2)	(3)	(4)	(5)	(6)
Hedef Grup*Post18	0.022*** (0.007)	0.020*** (0.007)	0.020*** (0.007)	-0.035** (0.014)	-0.021** (0.010)	-0.020** (0.010)
Year Fixed Effect	-	✓	✓	-	✓	✓
Region Fixed Effect	-	✓	✓	-	✓	✓
Regional Trend	-	-	✓	-	-	✓
Gözlem Sayısı	125,651	125,651	125,651	47,335	47,335	47,335

Note: The model uses observations from the HLFS dataset between 2016 and 2019. The Post18 variable takes the value of 1 for 2018 and 2019, and 0 for 2016 and 2017. The target group is defined as young men between the ages of 18 and 24. Estimates for the employment variable are obtained using a linear probability model. The wage variable is generated by taking the logarithm of the hourly real wage reported by individuals. The model includes dummy variables for education, age, and marital status, along with their interactions. The control group consists of male individuals between the ages of 25 and 28. Robust standard errors are clustered according to the NUTS2 regional classification and are given in parentheses. ***, **, and * indicate 1%, 5%, and 10% significance levels, respectively.

The impact of the employment program on the informal sector is presented in Table 6. The third and sixth columns report the estimation results of the model specified in Equation (4). According to these results, the program reduced the probability of young people being employed in the informal sector by 0.8 percentage points. On the other hand, the program's impact on real wages of young people in the informal sector is not statistically significant.

In the first and fourth columns, the model is estimated for employment and real wage variables, respectively, including only control variables. In the second and fifth columns, fixed effects are added to the control variables. For both employment and real wage outcomes, the fixed effects and regional trends do not appear to cause major changes in the results.

7.1.2. The Program Heterogeneity

When estimating the Average Treatment Effect (ATE), there is an implicit assumption that the treatment effect is similar across individuals with heterogeneous characteristics. However, individual characteristics can potentially alter the effect of a treatment on outcomes. This can result in the treatment effect differing from the average effect across various subgroups. The variation in a policy

Table 6. The Program Effect on Informal Employment

	Informal Sector					
	Employment			Real Wage		
	(1)	(2)	(3)	(4)	(5)	(6)
Target Group*Post18	-0.0095** (0.003)	-0.0095*** (0.003)	-0.0086** (0.003)	0.012 (0.029)	0.0068 (0.024)	0.0018 (0.024)
Year Fixed Effect	-	✓	✓	-	✓	✓
Region Fixed Effect	-	✓	✓	-	✓	✓
Regional Trend	-	-	✓	-	-	✓
Observation	125,651	125,651	125,651	10,207	10,207	10,207

Note: The model uses observations from the HLFS dataset between 2016 and 2019. The Post18 variable takes the value of 1 for 2018 and 2019, and 0 for 2016 and 2017. The target group is defined as young men between the ages of 18 and 24. Estimates for the employment variable are obtained using a linear probability model. The wage variable is generated by taking the logarithm of the hourly real wage reported by individuals. The model includes dummy variables for education, age, and marital status, along with their interactions. The control group consists of male individuals between the ages of 25 and 28. Robust standard errors are clustered according to the NUTS2 regional classification and are given in parentheses. ***, **, and * indicate 1%, 5%, and 10% significance levels, respectively.

or intervention's impact across different groups is referred to as Heterogeneous Treatment Effects (Cintron et al., 2022). Detecting heterogeneity in treatment effects is important, as it contributes to a clearer assessment of the policy outcomes and helps improve the design of future policies.

Heterogeneity in treatment effects may arise not only from individual characteristics but also from the period in which the intervention is implemented. In this section, the aim is to examine how the impact of the employment program varies both over time and across individuals with different levels of education. Additionally, the program's effect is estimated for various age groups. The objective here is to provide a clearer understanding of the program's overall impact.

7.1.2.1. The Effect of Program by Years

The impact of the employment program over the years was estimated using the model specified in Equation (5), and the results are presented in Table 7. According to the findings, the program had a greater impact on youth labor market outcomes in the year it was implemented. In 2018, the program increased the probability of young men being employed in the formal sector by 2.5 percentage points, whereas this figure declined to 1.6 percentage points in 2019. However, real wages in the formal sector for young people decreased by approximately 2% in 2018.

In the informal sector, the program's impact was only significant in the year it was launched. In 2018, the program reduced the probability of young people being employed in the informal sector by about 1 percentage point. For real wages in the informal sector, there is no statistically significant effect in either year.

Table 7. The Effect of Programs by Years

	Formal		Informal	
	(1) Employment	(2) Real Wage	(3) Employment	(4) Real Wage
2018	0.025*** (0.008)	-0.018* (0.009)	-0.011** (0.004)	0.017 (0.024)
2019	0.016** (0.007)	-0.022 (0.014)	-0.0062 (0.005)	-0.014 (0.032)
<i>Observation</i>	125,651	47,335	125,651	10,207

Note: The model uses observations from the HLFS dataset between 2016 and 2019. The target group is defined as young men aged 18-24. Estimates for the employment variable are obtained using a linear probability model. The wage variable is generated by taking the logarithm of the hourly real wage reported by individuals. The control group consists of male individuals aged 25-28. The model includes age, marital status dummies, and their interactions as control variables. Additionally, year, region fixed effects, and regional trend are used in all models. Robust standard errors are clustered according to the NUTS2 regional classification and are presented in parentheses. ***, **, and * indicate 1%, 5%, and 10% significance levels, respectively.

7.1.2.2. The Effect of Program by Education Levels

The impact of the employment program by education group is presented in Table 8. The program had no effect on the probability of being employed in the formal sector for individuals aged 18–24 who had no formal education, completed primary or middle school, or held a university degree. On the other hand, the program affected only high school graduates, increasing their probability of employment in the formal sector by 3.8 percentage points.

In the informal sector, the program was effective for individuals with no formal education and university graduates, reducing their probability of being employed in the informal sector by 5.3 and 1.4 percentage points, respectively.

Regarding real wages, the program had an effect on individuals employed in the formal sector who were primary/middle school and high school graduates. According to Table 8, the program reduced real wages for these groups by 2.5% and 6%, respectively. However, the program had no statistically significant effect on real wages in the informal sector for any education group.

7.1.2.3. The Effect of Program by Age Groups

The impact of the program by age group is presented in Table 9. According to the results, the program increased the probability of formal sector employment for young men aged 18–20 by 3 percentage points, while reducing their real wages by 3%. In the informal sector, the probability of employment for this age group decreased by approximately 1 percentage point. There is no significant effect on real wages for this group.

Table 8. The Effect of Program by Education Levels

	(1) No School	(2) Pri/ Sec School	(3) High School	(4) University
Formal Employment	-0.010 (0.026)	0.0010 (0.009)	0.038*** (0.011)	0.020 (0.016)
<i>Observation</i>	5,757	46,901	47,654	25,339
Formal Real Wage	0.018 (0.034)	-0.025** (0.011)	-0.060*** (0.010)	0.015 (0.021)
<i>Observation</i>	1,007	17,785	15,092	13,451
Informal Employment	-0.053** (0.024)	-0.0029 (0.008)	-0.0051 (0.005)	-0.014** (0.005)
<i>Observation</i>	5,757	46,901	47,654	25,339
Informal Real Wage	0.057 (0.061)	-0.0068 (0.025)	-0.076 (0.060)	0.058 (0.106)
<i>Observation</i>	1,013	5,726	2,723	745

Note: The model uses observations from the HLFS dataset between 2016 and 2019. The Post18 variable takes the value of 1 for the years 2018 and 2019 and 0 for the years 2016 and 2017. The target group is defined as young men between the ages of 18 and 24. The control group consists of men between the ages of 25 and 28. The model includes age, marital status dummies, and their interactions as control variables. Additionally, year, region fixed effects, and regional trend are used in all models. Robust standard errors are clustered according to the NUTS2 region classification and are given in parentheses. ***, **, and * indicate 1%, 5%, and 10% significance levels, respectively.

According to Table 9, the subsidy support had no effect on either the probability of formal sector employment or real wages for young men aged 21–24. However, it did reduce their probability of employment in the informal sector by 0.8 percentage points.

Table 9. The Effect of Program by Age Groups

	Formal		Informal	
	(1) Employment	(2) Real Wage	(3) Employment	(4) Real Wage
Male aged 18-20	0.030*** (0.009)	-0.030* (0.014)	-0.0093** (0.005)	0.014 (0.027)
<i>Observation</i>	81,482	30,846	81,482	6,723
Male aged 21-24	0.012 (0.007)	-0.015 (0.010)	-0.0080** (0.003)	-0.012 (0.027)
<i>Observation</i>	88,559	40,751	88,559	6,167

Note: The model uses observations from the HLFS dataset between 2017 and 2019. The Post18 variable takes the value of 1 for 2018 and 2019, and 0 for 2017. The control group consists of male individuals aged 25-28. Estimations for the employment variable were obtained using a linear probability model. The wage variable was created by taking the logarithm of the hourly wage reported by individuals. The model includes education, age, marital status dummies, and their interactions as control variables. Additionally, year and region fixed effects and the regional trend were used in all models. Robust standard errors are clustered according to the NUTS2 regional classification and are given in parentheses. ***, **, and * indicate 1%, 5%, and 10% significance levels, respectively.

Additional estimates for males aged 18–20 indicate that the program increased the probability of formal sector employment by 5 and 8.5 percentage points for high school and university graduates, respectively. However, there was a decline in real wages for primary/middle school and high school graduates by 4.3% and 5.4%, respectively. In the informal sector, the program did not have any statistically significant effect for this age group. The estimates by education level for the 18–20 age group are presented in Appendix 6.

For males aged 21–24, the additional estimates show that the effect of the program varied by education group. According to Appendix 7, the program affected only high school graduates in this age group, increasing their probability of employment in the formal sector by 2.4 percentage points. Similarly, only high school graduates experienced a decrease in real wages in the formal sector, by 6%. In the informal sector, the probability of employment decreased by 6 percentage points for those with no formal education and by 1.4 percentage points for university graduates. There is no statistically significant effect on real wages in the informal sector for any group within this age range.

7.1.3. Robustness Check

To assess the validity of the Difference-in-Differences estimates, a placebo test was first conducted. The placebo test plays an important role in identifying potential issues that may arise when estimating treatment effects in observational studies—such as confounding factors, model misspecification, or measurement error. Therefore, it is considered a strategy to check the validity of the findings and, more broadly, to strengthen causal inference (Eggers et al., 2021).

For the placebo test, the dataset was restricted to the pre-intervention period, and the model specified in Equation (4) was estimated under the assumption that the program was implemented in 2017. If there is no true treatment effect, the results should also appear significant when a fake intervention is introduced. The findings from the placebo test are presented in Table 10. There was no statistically significant change in either the employment probability or real wages of young individuals compared to the control group. These findings from the placebo test support the validity of the identification strategy.

In this section, the sensitivity of the Difference-in-Differences estimates over time is also examined by changing the observation window and re-estimating the effects. For this purpose, the sample was first restricted to cover the years 2016–2018, and the impact of the program on young men aged 18–24 was estimated. The results obtained are presented in Table 11 and are very similar to the

Table 10. Plasebo Test

	Formal		Informal	
	(1) Employment	(2) Real Wage	(3) Employment	(4) Real Wage
Target Group * Post17	0.0011 (0.009)	-0.010 (0.013)	0.0028 (0.005)	-0.020 (0.032)
<i>Observation</i>	63,897	24,522	63,897	5,382

Note: The model uses observations from the HLFS dataset between 2016 and 2017. The Post17 variable takes the value 0 for 2016 and 1 for 2017. The target group is defined as young men between the ages of 18 and 24. The control group consists of male individuals between the ages of 25 and 28. Estimates for the employment variable were obtained using a linear probability model. The wage variable was generated by taking the logarithm of the hourly wage reported by individuals. The model includes education, age, marital status dummies, and their interactions as control variables. Additionally, year and region fixed effects and the regional trend were used in all models. Robust standard errors are clustered according to the NUTS2 regional classification and are given in parentheses. ***, **, and * indicate 1%, 5%, and 10% significance levels, respectively.

overall effects of the program. Therefore, the program's sensitivity to the choice of years is quite low.

Table 11. The Effect of Program (2016-2018)

	Formal		Informal	
	(1) Employment	(2) Real Wage	(3) Employment	(4) Real Wage
Target Group * Post18	0.025** (0.011)	-0.022* (0.011)	-0.0099* (0.006)	0.00041 (0.028)
<i>Observation</i>	63,174	24,327	63,174	5,125

Note: The model uses observations from the HLFS dataset between 2016 and 2018. The Post18 variable takes the value 0 for 2016 and 2017, and 1 for 2018. The target group is defined as young men aged 18-24. The control group consists of men aged 25-28. Estimations for the employment variable were obtained using a linear probability model. The wage variable was generated by taking the logarithm of the hourly real wage reported by individuals. The model includes education, age, marital status dummies, and their interactions as control variables. Additionally, year and region fixed effects and regional trends were used in all models. Robust standard errors are clustered according to the NUTS2 regional classification and are given in parentheses. ***, **, and * indicate 1%, 5%, and 10% significance levels, respectively.

Another application to examine the sensitivity of the program's effect over time involved restricting the sample to include only the years 2017 and 2018 and estimating the causal effect. The results are presented in Table 12. These results are quite similar to the overall program effects for all variables except for real wages in the formal sector.

The estimates in Table 12 serve not only to demonstrate the program's sensitivity but also another purpose. Before the 2018 Additional Employment Incentive took its final form, there was a support program implemented in 2017 that covered all employees. However, this support was not in effect in 2016. There could be a very small effect during these years, but the results indicate that such an

effect is quite minimal.

Table 12. The Effect of Program (2017-2018)

	Formal		Informal	
	(1) Employment	(2) Real Wage	(3) Employment	(4) Real Wage
Target Group * 2018	0.024*** (0.008)	-0.013 (0.011)	-0.012*** (0.004)	0.030 (0.029)
<i>Observation</i>	62,507	24,043	62,507	5,205

Note: The model uses observations from the HLFS dataset between 2017 and 2018. The target group is defined as young men aged 18-24. The control group is comprised of men aged 25-28. Estimates for the employment variable are obtained using a linear probability model. The wage variable is generated by taking the logarithm of the hourly real wage reported by individuals. The model includes education, age, marital status dummies, and their interactions as control variables. Additionally, year and region fixed effects and regional trends are used in all models. Robust standard errors are clustered according to the NUTS2 regional classification and are presented in parentheses. ***, **, and * indicate 1%, 5%, and 10% significance levels, respectively.

7.2. REGRESSION DISCONTINUITY DESIGN ESTIMATION RESULTS

This section presents the estimates obtained using the Regression Discontinuity Design (RDD). Similar to the DiD method, the estimates were made for employment and real wage variables in both the formal and informal sectors. Equation (10) was estimated for the linear model, and Equation (11) for the quadratic model. The estimates obtained using RDD for various bandwidths are shown in Table 13. Panel A includes results for males aged 23–26, Panel B for ages 22–27, and Panel C for ages 21–28. All estimates were conducted separately for the years 2018 and 2019. Additionally, a placebo test was performed for 2017 to examine the reliability of the estimates.

As shown in Panel A, for males aged 23–26, the estimates for 2017 based on both the linear and quadratic models show no statistically significant effect of the program on employment and real wages in the formal sector. However, the program’s effect on employment in the informal sector is statistically significant. Therefore, the informal employment variable fails the placebo test. According to the estimates for males aged 23–26, the program increased the probability of formal sector employment by 6.3 percentage points in the linear model and by 4.2 percentage points in the quadratic model for 2018. Although negative estimates for informal sector employment were obtained, these results are not statistically significant. In 2019, the linear model shows no statistically significant effects for employment variables, indicating that, unlike the Difference-in-Differences estimates, the program’s effect was limited to the year it was implemented (2018). Regarding real wages in the formal sector, no statistically significant effect was found for 2018 or 2019. In the informal sector, however, very large and statistically significant parameter estimates were obtained

only in 2019. Specifically, the average real wages of males aged 23–24 in the treatment group were 22% lower than those in the control group in 2019. The absence of any significant effect on employment in both sectors for 2019 complicates the interpretation of this finding on real wages. Moreover, the low number of observations in these estimates raises questions about their reliability. According to Panel B, for males aged 22–27, the program’s effect in 2017 is statistically significant only for informal employment in the linear model. Thus, the informal employment variable fails the placebo test for this age group in the linear model. In the quadratic model, there is no statistically significant effect of the program on any variables for 2017. For 2018, the program increased the probability of employment by 10 percentage points in the quadratic model for this age group. No such effect was observed in the linear model for employment variables. Additionally, according to the linear model, the program reduced real wages in 2019 by 1.7% in the formal sector and by 12% in the informal sector. The quadratic model shows that the program reduced real wages in the informal sector only in 2019.

Panel C shows that for males aged 21–28, the program affected only informal employment in 2017 in the linear model, so this variable fails the placebo test. For this age group, the program’s effects on employment variables were observed only in 2018. According to Table 13, the program increased the probability of formal sector employment by 3.3 percentage points in the linear model in 2018. In 2019, the program reduced real wages in the formal sector by 3.4%. In the quadratic model, the program was found to affect only real wages in the informal sector.

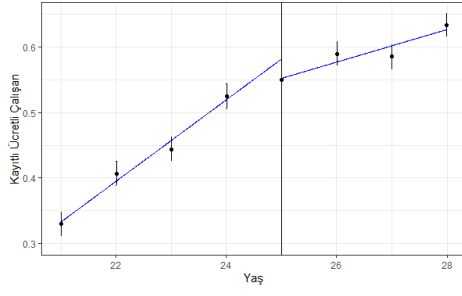
The estimates obtained using different bandwidths show that the RDD estimates are quite sensitive. Furthermore, the model developed for estimating this method was estimated with control variables added for all bandwidths, and the results are presented in Appendix 13. While similar results were obtained for males aged 23–26 and 21–28, some differences were observed in the estimates for males aged 22–27. Therefore, it was also found that the estimates are sensitive to the inclusion of control variables.

The graphical representation of the causal effect obtained by the Regression Discontinuity Design for the year 2018 is presented in Figure 8 for the linear model and in Figure 9 for the quadratic model.

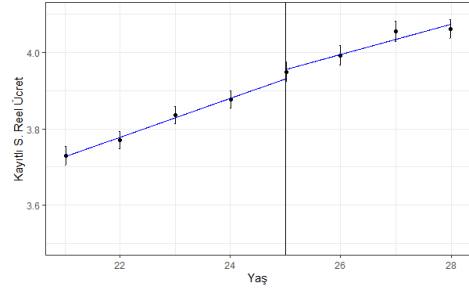
Table 13. Regression Discontinuity Method Estimation Results

	Formal		Informal	
	(1) Employment	(2) Real Wage	(3) Employment	(4) Real Wage
Panel A: Males Aged 23-26 (Bandwidth: 2 Years)				
<i>Linear Model</i>				
2017	-0.022 (0.022)	-0.003 (0.030)	0.026* (0.014)	0.102 (0.076)
Observation	11,298	5,369	11,298	884
2018	0.063*** (0.023)	-0.028 (0.030)	-0.017 (0.013)	-0.057 (0.084)
Observation	10,902	5,281	10,902	733
2019	0.010 (0.023)	-0.012 (0.033)	-0.006 (0.013)	-0.219** (0.092)
Observation	11,069	4,963	11,069	701
<i>Quadratic Model</i>				
2017	-0.010 (0.021)	-0.036 (0.027)	0.022* (0.013)	0.035 (0.073)
Observation	11,298	5,369	11,298	884
2018	0.042** (0.021)	-0.024 (0.027)	-0.016 (0.012)	0.005 (0.077)
Observation	10,902	5,281	10,902	733
2019	0.003 (0.021)	-0.042 (0.029)	-0.007 (0.012)	-0.223*** (0.084)
Observation	11,069	4,963	11,069	701
Panel B: Males Aged 22-27 (Bandwidth: 3 Years)				
<i>Linear Model</i>				
2017	-0.002 (0.016)	-0.003 (0.022)	0.019* (0.010)	0.053 (0.059)
Observation	16,972	8,069	16,972	1,283
2018	0.024 (0.017)	-0.006 (0.022)	-0.002 (0.010)	-0.022 (0.063)
Observation	16,235	7,731	16,235	1,108
2019	0.025 (0.016)	-0.019 (0.024)	-0.017* (0.010)	-0.119* (0.064)
Observation	16,377	7,171	16,377	1,079
<i>Quadratic Model</i>				
2017	-0.050 (0.041)	0.012 (0.057)	0.036 (0.025)	0.174 (0.144)
Observation	16,972	8,069	16,972	1,283
2018	0.109*** (0.042)	-0.063 (0.058)	-0.033 (0.024)	-0.190 (0.155)
Observation	16,235	7,731	16,235	1,108
2019	-0.019 (0.041)	-0.012 (0.062)	0.009 (0.024)	-0.311* (0.159)
Observation	16,377	7,171	16,377	1,079
Panel C: Males Aged 21-28 (Bandwidth: 4 Years)				
<i>Doğrusal Model</i>				
2017	-0.001 (0.014)	-0.017 (0.019)	0.023*** (0.008)	0.029 (0.051)
Observation	22,292	10,483	22,292	1,637
2018	0.033** (0.014)	-0.017 (0.019)	-0.007 (0.008)	-0.011 (0.052)
Observation	21,480	10,093	21,480	1,466
2019	0.009 (0.014)	-0.034* (0.020)	-0.009 (0.008)	-0.083 (0.055)
Observation	21,756	9,373	21,756	1,462
<i>Panel C: Kuadratik Model</i>				
2017	-0.031 (0.027)	0.021 (0.038)	0.017 (0.016)	0.109 (0.097)
Observation	22,292	10,483	22,292	1,637
2018	0.040 (0.027)	-0.015 (0.039)	-0.005 (0.016)	-0.051 (0.104)
Observation	21,480	10,093	21,480	1,466
2019	0.033 (0.027)	0.016 (0.042)	-0.019 (0.016)	-0.254** (0.106)
Observation	21,756	9,373	21,756	1,462

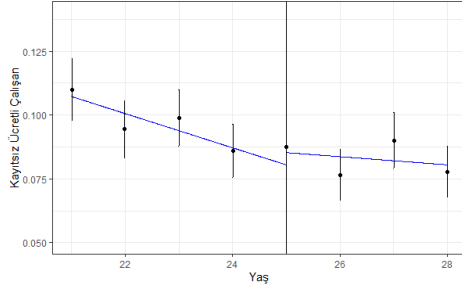
Not: The cutoff point was set at 25. Estimates for the employment variable were obtained using a linear probability model. The wage variable was created by taking the logarithm of the hourly real wage reported by individuals. A region fixed effect was used in all models. Robust standard errors are clustered according to the NUTS2 region classification and are given in parentheses. ***, **, and * indicate 1%, 5%, and 10% significance levels, respectively.



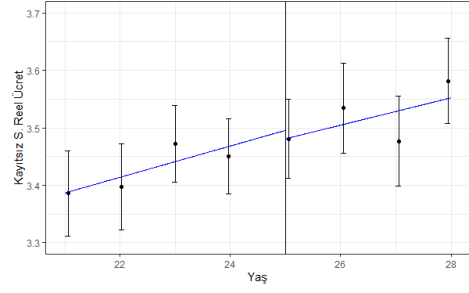
Formal Employment



Formal Real Wage

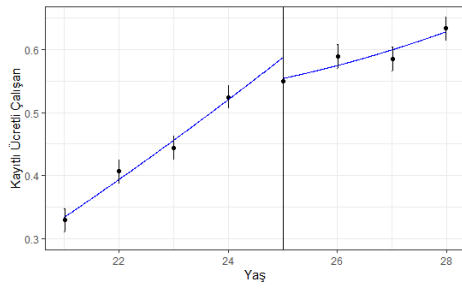


Informal Employment

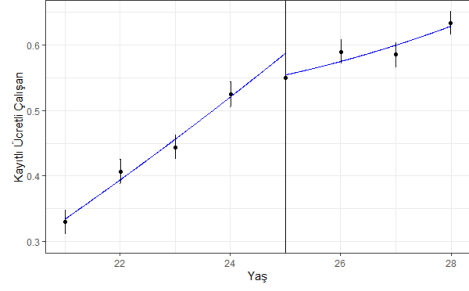


Informal Real Wage

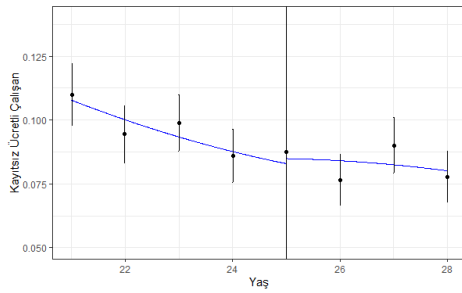
Figure 7. Regression Discontinuity Method Graphs (Linear Model)- 2018



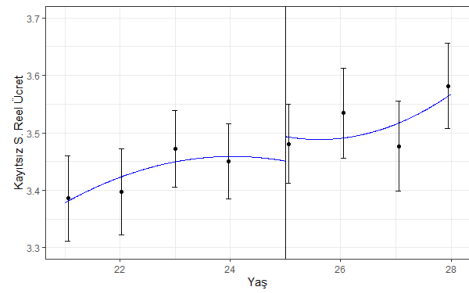
Formal Employment



Formal Real Wage



Informal Employment



Informal Real Wage

Figure 8. Regression Discontinuity Method Graphs (Quadratic Model)- 2018

CHAPTER VIII. CONCLUSION

Unemployment is a global issue faced by all countries around the world. However, what sets Turkey apart from other countries is the significantly high rate of youth unemployment. Considering that a large portion of Turkey's population consists of young people, it is clear that youth unemployment will lead to serious problems in the future. Active labor market policies are one of the tools that governments use to increase the labor force participation and employment of relatively disadvantaged groups in the labor market, such as women, youth, migrants, or people with disabilities. Under the Additional Employment Program introduced in 2018, the extension of insurance premium support provided to employers on the condition of employing young men over the age of 18 and under the age of 25 allows this incentive package to be considered as an active labor market policy. This study aims to analyze the effect of this program on labor market outcomes for young men using the Difference-in-Differences (DiD) method and the Regression Discontinuity Design (RDD). By analyzing a specific active labor market program for the Turkish labor market, this thesis makes a new contribution to the literature. Furthermore, the fact that the program was not implemented during a period of crisis enables a more straightforward measurement of its effects.

While estimating the effect of the program, efforts were made to isolate the impact from other simultaneous subsidy packages implemented in Turkey. For this purpose, in the Difference-in-Differences analysis, the control group was selected from males aged 25–28. The treatment group consisted of males aged 18–24. Separate estimations were conducted for different years, education levels, and age groups to identify potential heterogeneous effects of the program across groups and time. Since the program was designed based on age eligibility, both the DiD and RDD methods could be applied in the analysis.

According to the estimates from the Difference-in-Differences method, the employment program increased the probability of employment in the formal sector among young men by 2 percentage points, while reducing the probability of informal sector employment by approximately 1 point. Additionally, the subsidy package reduced hourly real wages in the formal sector by an average of 2%. The increase in formal employment coupled with a decrease in real wages may be explained

by the employment of younger and less educated workers with limited experience. The increase in formal employment likelihood is expected to encourage young men previously working in the informal sector to transition into formal employment. Indeed, the program appears to have reduced the probability of informal employment among young men. Moreover, the impact of the program on employment is observed to diminish over time. The largest effect on formal employment was seen in 2018, the year the program was launched, with an increase of 2.5 percentage points. In 2019, the effect was estimated at 1.6 points. However, the impact on informal employment and real wages seems to have disappeared by 2019.

When the analysis was repeated for subgroups based on educational attainment, the program was found to affect only high school graduates, increasing both their probability of formal sector employment and their real wages. The program reduced the likelihood of informal employment among individuals with no school completion and university graduates.

Several robustness checks were performed to test the reliability of the Difference-in-Differences estimates. The results from these tests confirm that the estimates obtained using the DiD method are robust and reliable.

According to the Regression Discontinuity Design, the program only affected youth labor market outcomes in the year it was implemented. When using the narrowest bandwidth and a linear model, the program increased the probability of formal sector employment among young men by 6 percentage points in 2018; under a quadratic model, the increase was 4 points. On the other hand, no significant effect of the program on the likelihood of informal employment among young men was found. The RDD estimates for real wages yield negative and statistically significant coefficients. Given that the DiD and RDD rely on different assumptions, it is expected that the estimates from these two methods differ. Since the DiD method estimates the average treatment effect on a broader group, it typically yields smaller parameter estimates than RDD. RDD estimates were also found to be highly sensitive to the chosen age intervals. Moreover, estimates varied when control variables were added to the model. Therefore, we conclude that the DiD estimates provide a more reliable assessment.

The impact of the 2018 Additional Employment Program's regulation targeting males aged 18–24 on formal sector employment is comparable in magnitude to those reported in the existing literature. In particular, the effect on the formal employment of high school graduates—where the main impact occurred—was approximately 3.8 percentage points. In the relevant literature, estimates

for the 2008 Employment Support Program in Turkey ranged between 0.5 and 5 percentage points (Ayhan, 2013; Balkan et al., 2016b; Özdamar et al., 2021). Considering that the 2018 regulation extended the duration of eligibility for the existing Additional Employment Program, it is reasonable that its impact may have been smaller than the 2008 intervention.

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APPENDICES

Appendix 1. Variable Definitions

Formal Employment: A dummy variable taking the value 1 if the individual works as a wage employee registered with the social security institution, and 0 otherwise.

Informal Employment: A dummy variable taking the value 1 if the individual works as a wage employee not registered with the social security institution, and 0 otherwise.

Real Wages in Formal and Informal Sectors: When calculating the hourly real wages of individuals employed as wage workers in the formal and informal sectors, the income level reported by the individuals is first divided by the inflation rate of that year. This yields the individual's real monthly wage. Then, the weekly working hours reported by the individual are multiplied by 4.2 to find the total hours worked per month. Finally, the monthly real wage is divided by the monthly working hours to calculate the hourly real wage. The logarithm of the hourly real wage is taken when performing estimations.

Education Levels: In the HIA dataset, individuals' education status is indicated in seven categories: primary school, general middle school, vocational or technical middle school, general high school, vocational or technical high school, 2- or 3-year vocational college, 4-year university or faculty, master's degree (including 5- or 6-year faculties), or doctorate. In this thesis, education levels are grouped into four categories: individuals with no formal education, primary or middle school graduates, high school graduates, and finally university graduates (undergraduate and graduate).

Marital Status: A dummy variable taking the value 1 if the individual is married, and 0 otherwise.

Post18: A dummy variable taking the value 1 for the years 2018 and 2019, and 0 for 2016 and 2017.

Target Group: A dummy variable taking the value 1 if the individual is between 18 and 24 years old, and 0 otherwise.

The codes used for the analyses in this thesis can be accessed at the following link: https://github.com/yaseminhayirli/master_thesis

Please contact us for the tables in the appendix.